MAE3405 Aerospace Propulsion

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Topic 7-8 Problem Set Questions

An asterisk indicates that this problem is similar to a past exam question.

Question 16*

A 3-bladed propeller is being statically tested at SSL with zero advance speed at 2700RPM. The blade length is 1.0m from centre to tip; the blade hub is 0.05m. The blade chord varies as $c = 0.1R\left(1.1 - \frac{r}{R}\right)$. The angle of attack of the blade varies with blade radius according to the relationship $\alpha = 12 - 4\left(\frac{r}{R}\right)$ [degrees] where r is the radius and R the blade length. The slope of the lift curve for the aerofoil section of the blade is equal to 0.15 deg^{-1}.

(i) Determine the thrust produced.

(ii) Estimate the induced velocity.

[Ans: (i) 2.4kN, (ii) 18m/s.]

Question 17*

An intercooled turbocharged 4 cylinder 4-stroke spark ignition aircraft piston engine fuelled by avgas drives a 1.8m diameter twin blade propeller. Data on the engine is listed in the table below. The turbocharger provides a pressure ratio of 1.3 at an isentropic efficiency of 85%. It is located between the intake and fuel injection system. The turbocharger's intercooler reduces the turbocharger temperature rise by 75%. Assuming the engine to operate on the fuel/air Otto cycle, calculate the static thrust available at SSL using Froude theory.

Piston Engine			
Quantity	**Value**	**Quantity**	**Value**
Compression ratio	7.0	Combustion efficiency	95%
Bore	0.1m	Mechanical Efficiency	90%
Stroke	0.12m	Volumetric efficiency	100%
RPM	2500	Air to fuel ratio	14.5
Cp (fuel & air)	1.055kJ/kg-K	Cp (exhaust products)	1.486kJ/kg-K
Cv (fuel & air)	0.768kJ/kg-K	Cv (exhaust products)	1.199kJ/kg-K

[Ans: 4.4 kN]

Question 17C*

The Russian Mil Mi-6 heavy lift helicopter is powered by two turboshaft engines with SSL rate power of 4050 kW each. Without a payload, the take-off mass is 28,000 kg. The 5-bladed main rotor has a diameter of 35 m. If an amount equal to 20% of the SSL rated engine power is required to drive the tail rotor and aircraft accessories at any power setting, determine:

(i) The maximum SSL level payload that can just be lifted from the ground

(ii) The maximum altitude the helicopter can hover without a payload.

[Ans: (i) 19,153 kg. (ii) 4100 m].

Question 19

A helicopter with total mass 600 kg, drag coefficient 0.35 and rotor diameter 10 m, moves with a forward speed of 25 knots at an altitude of 1000m. By determining the relative speed of the air to the blade, find the power required.

[Ans: 49.16 kW]